
ESPACE: Advanced Orbital Mechanics

Exercise: Homogeneous Solution of Hill Equations

Satellite orbits are disturbed by the Earth gravity field. For (near) circular orbits these disturbances can be approximately computed using the analytical Hill theory. As for satellites in close formation perturbations are very similar, the perturbations nearly cancel in the relative motion of the satellites. The homogeneous Hill equations can thus be used approximately represent the relative motion.

For the lab the following numerically integrated TerraSAR and TanDEM-X orbit vectors are obtained from DLR in the inertial system. The files terrasarsar.zip and tandemx.zip, which can be downloaded from the Moodle web site, contain six columns of data ready to be read with Matlab. The first three columns give the components of the satellite position vector in inertial frame, the last three columns give the components of the velocity vector in inertial frame. Units are m and m/s. Step size is 10 sec.

Tasks:

1. Display the orbits of the two satellites. Plot the components of the difference vector between the two satellites as a function of time. Plot the distance between the two satellites as well as the absolute value of the velocity difference and the radial velocity.
2. Define the orbit-fixed reference system related to TerraSAR (z: radial direction; y: crosstrack direction; x: perpendicular to the two in direction of satellite motion; origin: position of satellite). Plot the components of the velocity vector of TerraSAR in this reference frame and discuss the result.
3. Display the position difference and the velocity difference vector to TanDEM-X in the same reference frame. Display the motion of TanDEM-X around TerraSAR.

In the following exercises the homogeneous Hill equations are used to represent the nearly unperturbed motion of TanDEM-X around TerraSAR. For simplicity we assume that the eccentricity of the TerraSAR orbit is small enough such that the conditions for the Hill equations are approximately fulfilled. The goal is to assess the difference between the perturbed and the unperturbed motion of TanDEM-X.

4. Determine the mean motion of TerraSAR from the orbit, e.g., by associating the mean geocentric distance with the semimajor axis and using Kepler's third law.
5. The solution of the homogeneous Hill equations contain six initial conditions. Determine these initial position and velocity vectors of TanDEM-X relative to TerraSAR in an inverse calculation, i.e., adjust the Hill solution to the true relative orbit with a Least Squares procedure (see equations below). Use the satellite positions between the two manoeuvres at epoch numbers 2000 and 9000 for this purpose.
6. Display the corresponding solution of the Hill equation with the initial conditions obtained from the previous step. Compare this orbit with the perturbed relative orbit.

Please prepare a short report and comment the results.

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Formula Sheet

Homogeneous Solution of Hill Equations

The formulas on this page represent a summary of all required formulas for the exercise.

Hill Equations: In general form they are defined as:

$$\ddot{x} + 2\bar{n}\dot{z} = g_x$$

$$\ddot{y} + \bar{n}^2 y = g_y$$

$$\ddot{z} - 2\bar{n}\dot{x} - 3\bar{n}^2 z = g_z$$

With $\bar{n}$ for the mean motion of a satellite on a circular orbit and (x,y,z) the coordinates in the local orbit system.

Forward Computations: If we approximate the Earth as a sphere, the derivatives of the disturbing potential become zero and we get as solution of the homogeneous linear differential equations the following equations:

$$x(t) = \frac{2}{\bar{n}} \dot{z}_0 \cos \bar{n}t + \left(\frac{4}{\bar{n}} \dot{x}_0 + 6z_0 \right) \sin \bar{n}t - (3\dot{x}_0 + 6\bar{n}z_0)t + x_0 - \frac{2}{\bar{n}} \dot{z}_0$$

$$y(t) = y_0 \cos \bar{n}t + \frac{\dot{y}_0}{\bar{n}} \sin \bar{n}t$$

$$z(t) = \left(-\frac{2}{\bar{n}} \dot{x}_0 - 3z_0 \right) \cos \bar{n}t + \frac{\dot{z}_0}{\bar{n}} \sin \bar{n}t + \frac{2}{\bar{n}} \dot{x}_0 + 4z_0$$

With the initial conditions $\mathbf{x}(t=0) = \mathbf{x}_0$ and $\dot{\mathbf{x}}(t=0) = \dot{\mathbf{x}}_0$. These equations are applied for the forward computation to generate a time series of position vectors and after differentiation with respect to time to generate a time series of velocity vectors.

$$\dot{x}(t) = -2\dot{z}_0 \sin \bar{n}t + (4\dot{x}_0 + 6\bar{n}z_0) \cos \bar{n}t - 3\dot{x}_0 - 6\bar{n}z_0$$

$$\dot{y}(t) = -y_0 \bar{n} \sin \bar{n}t + \dot{y}_0 \cos \bar{n}t$$

$$\dot{z}(t) = (2\dot{x}_0 + 3\bar{n}z_0) \sin \bar{n}t + \dot{z}_0 \cos \bar{n}t$$

Inverse Computations: From inversion of the forward computations we can determine the initial conditions from a time series of orbit positions. The y-component can be handled separately, while x and z are coupled. It holds (Least Squares procedure):

$$\begin{pmatrix} x_0, \dot{x}_0, z_0, \dot{z}_0 \end{pmatrix}^T = \left(\mathbf{A}_{xz}^T \mathbf{A}_{xz} \right)^{-1} \mathbf{A}_{xz}^T \begin{pmatrix} x, z \end{pmatrix}^T$$

and

$$\begin{pmatrix} y_0, \dot{y}_0 \end{pmatrix} = \left(\mathbf{A}_y^T \mathbf{A}_y \right)^{-1} \mathbf{A}_y^T y$$

where the A-matrices are determined by the partial derivatives with respect to the initial values. MATLAB tip: Solution of a system of linear equations: $y = \mathbf{A}x : x = \mathbf{A} \backslash y$

In particular the partial derivatives are defined by:

$$\begin{aligned}
\frac{\partial x}{\partial x_0} &= 1 \\
\frac{\partial x}{\partial \dot{x}_0} &= \frac{4}{\bar{n}} \sin \bar{n}t - 3t \\
\frac{\partial x}{\partial z_0} &= 6(\sin \bar{n}t - \bar{n}t) \\
\frac{\partial x}{\partial \dot{z}_0} &= \frac{2}{\bar{n}} (\cos \bar{n}t - 1) \\
\frac{\partial z}{\partial x_0} &= 0 \\
\frac{\partial z}{\partial \dot{x}_0} &= \frac{2}{\bar{n}} (1 - \cos \bar{n}t) \\
\frac{\partial z}{\partial z_0} &= 4 - 3 \cos \bar{n}t \\
\frac{\partial z}{\partial \dot{z}_0} &= \frac{1}{\bar{n}} \sin \bar{n}t \\
\frac{\partial y}{\partial y_0} &= \cos \bar{n}t \\
\frac{\partial y}{\partial \dot{y}_0} &= \frac{1}{\bar{n}} \sin \bar{n}t
\end{aligned}$$

The resulting A-matrices then are defined by:

$$\mathbf{A}_{xz} = \begin{pmatrix} \frac{\partial x}{\partial x_0} & \frac{\partial x}{\partial \dot{x}_0} & \frac{\partial x}{\partial z_0} & \frac{\partial x}{\partial \dot{z}_0} \\ \frac{\partial z}{\partial x_0} & \frac{\partial z}{\partial \dot{x}_0} & \frac{\partial z}{\partial z_0} & \frac{\partial z}{\partial \dot{z}_0} \end{pmatrix} \text{ and } \mathbf{A}_y = \begin{pmatrix} \frac{\partial y}{\partial y_0} & \frac{\partial y}{\partial \dot{y}_0} \end{pmatrix}$$